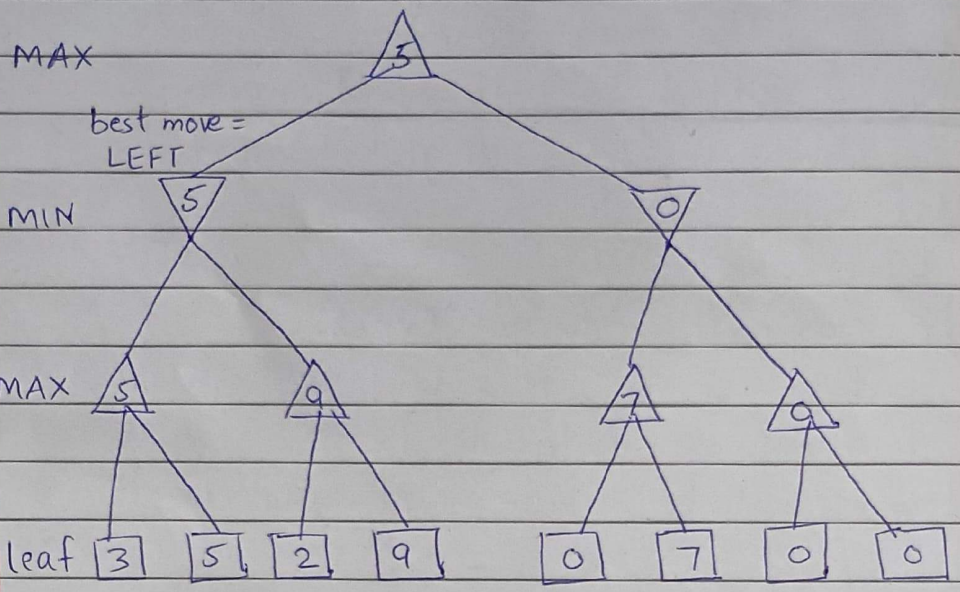
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Lab 5 - Minimax Trace (Task 1)

- Tree from code 5.1: scores = [3, 5, 2, 9, 0, 7] code pads it with 0s so it becomes a full tree: 8 leaves, depth 3



triangle UP = MAX node (picks the biggest)
triangle DOWN = MIN node (picks the smallest)
blue number = given score, red number = value 1 computed

Task 1: Bottom-up Calculation

level numbers: root = depth 0, leaves = depth 3
N = MAX nodes, M = MIN nodes

Level	Node	Children	Op	Value
3(MAX)	N1	3, 5	max	5
3(MAX)	N2	2, 9	max	9
3(MAX)	N3	0, 7	max	7
3(MAX)	N4	0, 0	max	0
1(MIN)	M1	5, 9	min	5
1(MIN)	M2	7, 0	min	0
root	R	5, 0	max	5

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Answer: optimal value = 5
Best action: go LEFT (to M1) because 5 > 0

Task 2: Leaf Counter

Idea: count a leaf every time the code reaches the base case.

What I added to both functions:

```
count = 0                    # global counter
def minimax(...):
global count
if curDepth == targetDepth:
count += 1                   # a leaf was evaluated
return scores[nodeIndex]
(same 2 lines inside alphabeta too)
```

Results (reset count = 0 before each run):

Tree	Minimax	leaves	AB	leaves
5.1 tree [3,5,2,9,0,7,0,0]	5	8	5	8
5.2 tree [3,5,2,9,3,5,2,9]	5	8	5	6

Same root value (5) both times, alpha-beta never uses MORE leaves.

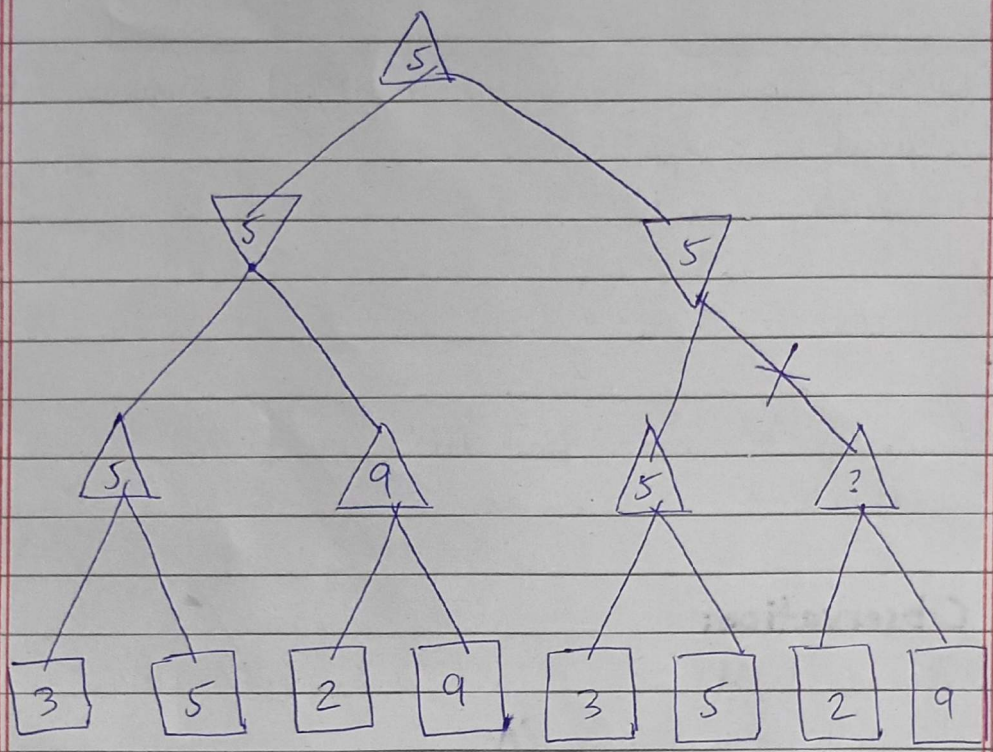
Why no pruning on the 5.1 tree?

The 0s sit on the right and are checked last. When the right MIN node gets its first value 7, alpha = 5 is not >= beta yet, so nothing can be cut. Pruning depends on the ORDER of leaves.

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Alpha-Beta Trace on 5·2 Tree

scores = [3, 5, 2, 9, 3, 5, 2, 9]



PRUNED: leaves 2, 9 never read

Step	Node	alpha	beta	What happens
1	R	-inf	+inf	go to M1, then N1
2	N1	-inf	+inf	Leaves 3, 5 → 5
3	M1	-inf	5	beta = 5, go to N2
4	N2	-inf	5	leaf 2, leaf 9 → 9
5	M1	-inf	5	min (5, 9) = 5
6	R	5	+inf	alpha = 5, go to M2
7	N3	5	+inf	leaves 3, 5 → 5
8	M2	5	5	alpha >= beta: CUT N4
9	R	5	+inf	max (5, 5) = 5

Leaves evaluated = 6 (3, 5, 2, 9, 3, 5) skipped = 2

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Task 3: Reorder the Branches

Same tree, I only swap the order of children. Minimax value must stay 5.

Order	leaves left to right	root	Minimax	AB
original	3 5 2 9 3 5 2 9	5	8	6
best order	5 3 9 2 5 3 9 2	5	8	5
worst order	2 9 3 5 2 9 3 5	5	8	8
mirrored	9 2 5 3 9 2 5 3	5	8	8

(columns Minimax and AB = number of leaves evaluated)

Observation:

- The root value is ALWAYS 5, pruning does not change the answer.
- Minimax always reads 8 leaves. Alpha-beta reads 5 to 8.
- Best case: only 5 leaves, 3 skipped (a 37% saving).

- Why is the best order good? (leaves 5 3 9 2 5 3 9 2)
  M1 first gets 5 from N 1, so beta = 5.
  In N2 the very first leaf is 9, and 9 >= 5 → leaf "2" is skipped.
  Root gets alpha = 5. In M2 the first MAX node gives 5, then alpha (5) >= beta (5) → the whole N4 (2 leaves) is skipped.
- Why is the worst order bad? (leaves 2 9 3 5 2 9 3 5)
  The best value is met LAST in every pair (2 then 9, 3 then 5), so alpha and beta never get tight early and nothing is cut. Same tree, same answer, but 8 leaves = no better than minimax.